

Spring Boot in the Cloud

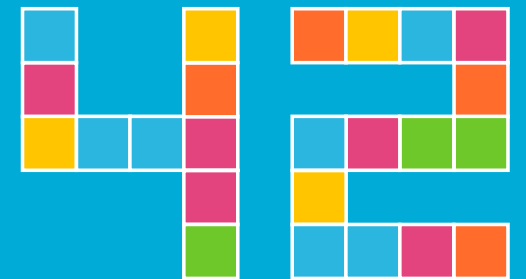
Advanced Optimization Deep Dive

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TALENTS

Abstract

Spring Boot in the Cloud: Advanced Optimization Deep Dive

Spring Boot makes it easy to launch cloud-ready applications, but building truly lean and efficient cloud services requires going beyond the basics. In this three-hour deep dive, you'll immerse yourself in advanced Spring Boot techniques to optimize your applications for the cloud.

We will examine in depth how to reduce memory usage, improve startup times, and streamline your Spring Boot apps for modern cloud environments. Topics include Spring AOT, classpath exclusions, lazy beans, actuator configuration, custom JVM options, and additional state-of-the-art tools.

This session combines short presentations, live coding, and practical hands-on exercises. You will experiment with advanced configurations, observe real-time results, and troubleshoot performance challenges alongside the instructor. Real-world scenarios and best practices will help you immediately apply these techniques to your own projects.

Whether you want to fine-tune your CI/CD pipeline or push Spring Boot to its limits in production, you will leave with actionable skills, a collection of code samples, and a deeper understanding of cloud-native application optimization. Get ready for an in-depth, practical exploration of building lean Spring Boot applications for the cloud.

Spring Boot in the Cloud

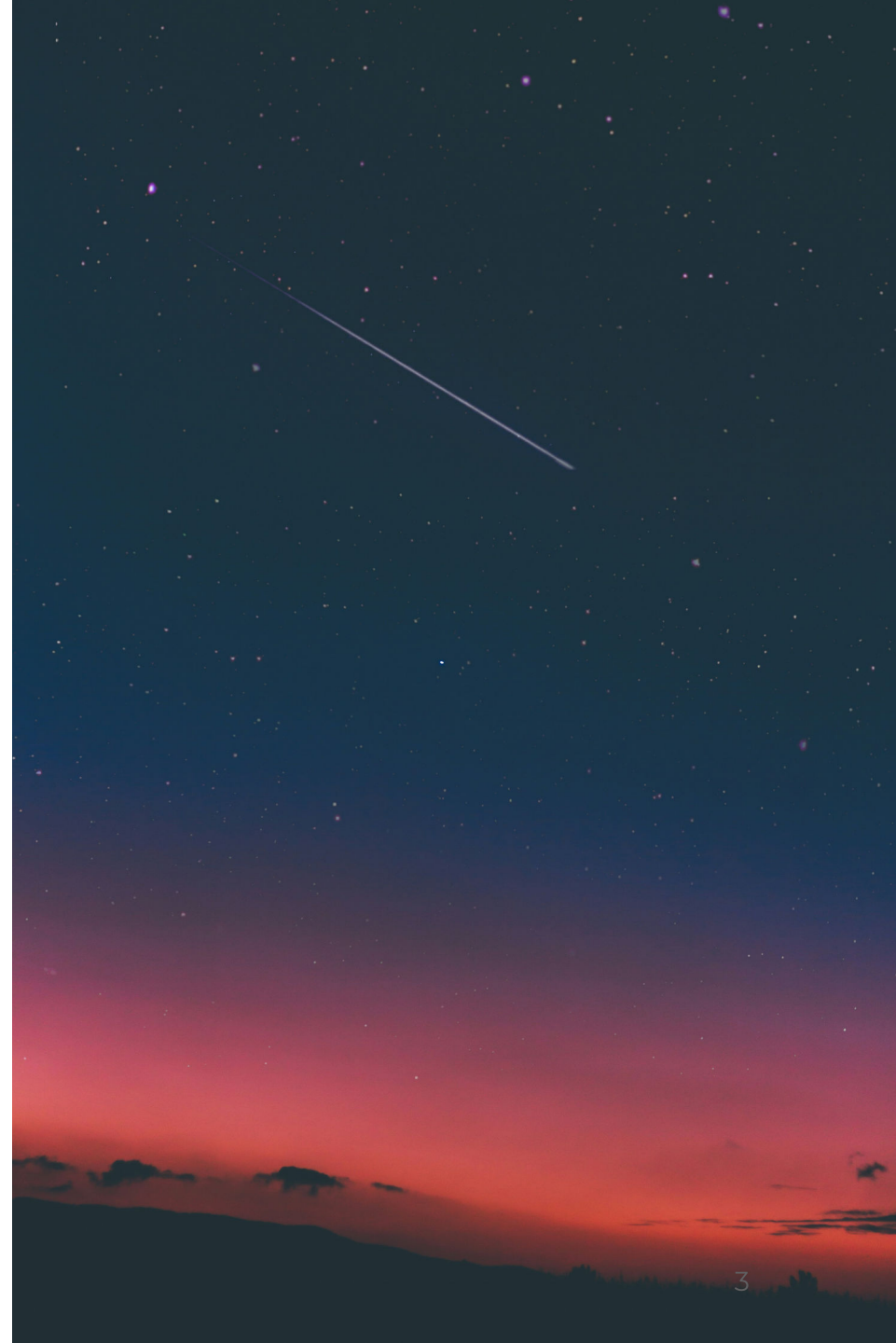
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Spring Boot in the Cloud

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bit.ly/48i5U9x



WARNING:

Numbers shown in this talk are **not** based on **real data** but **only estimates** and **assumptions** made by the **author** for **educational purposes** only.

Introduction



Patrick Baumgartner

Technical Agile Coach @ **42talents**

My focus is on the **development of software solutions *with* humans.**

Coaching, Architecture, Development, Reviews, and Training.

Lecturer @ **Zurich University of Applied Sciences ZHAW**

Co-Organizer of **Voxxed Days Zurich, JUG Switzerland**, Java Champion, Oracle ACE Pro Java ...

[@patbaumgartner](https://twitter.com/patbaumgartner)

What's the challenge?

Why does this matter?

I ❤️ Java 🤗 & Spring Boot 🌿



Introduction to Optimization Goals

Questions ...

Think about your current project.

- What would you optimise your application for?
- What are the key metrics?
- How would you measure them?

3' in pairs

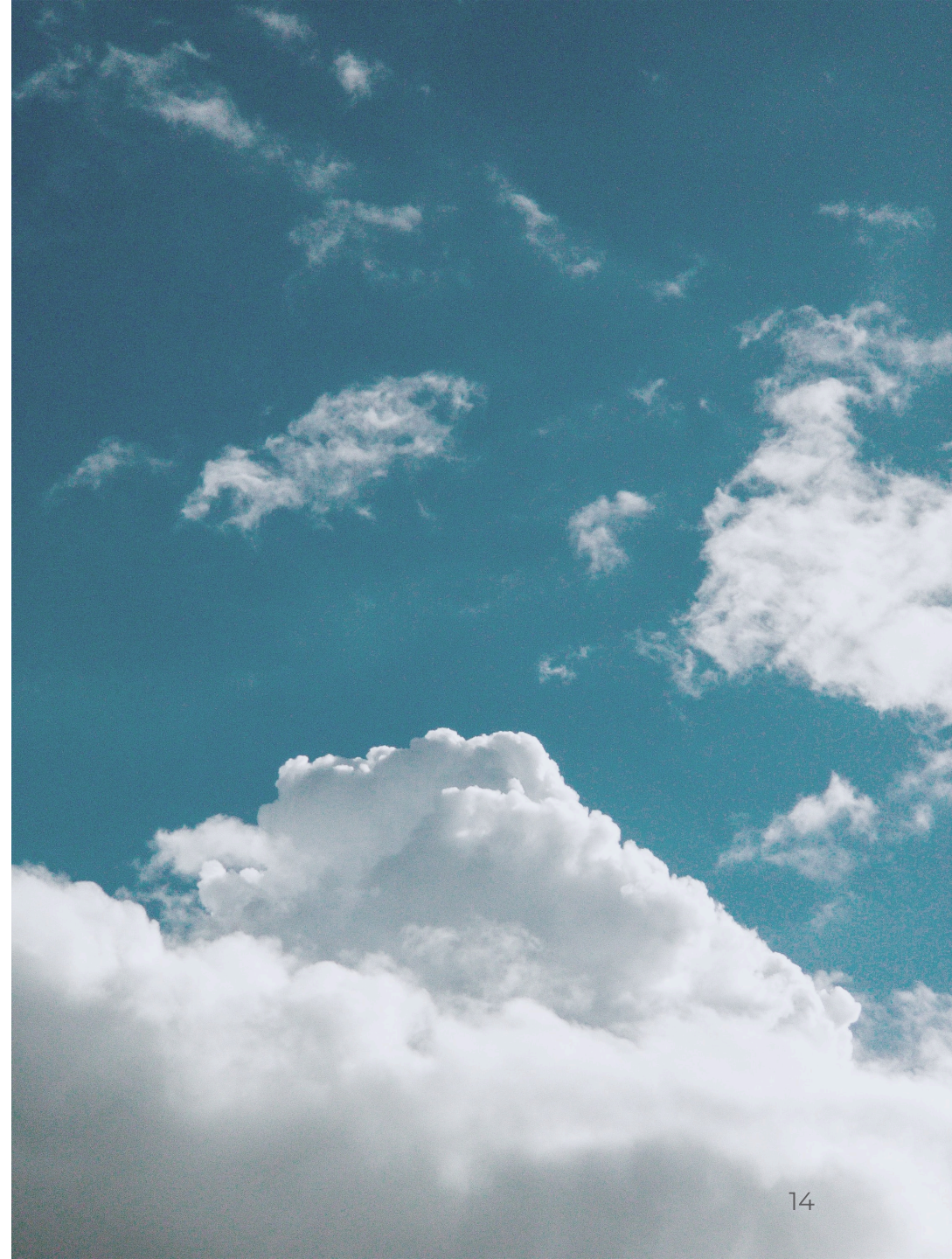
Measures

- Use Case
- Build Time
- Image Size
- Startup Time
- CPU Usage
- Memory Usage
- Throughput / Requests per Second
- Latency / Response Time

Requirements

When Deploying to a Cloud

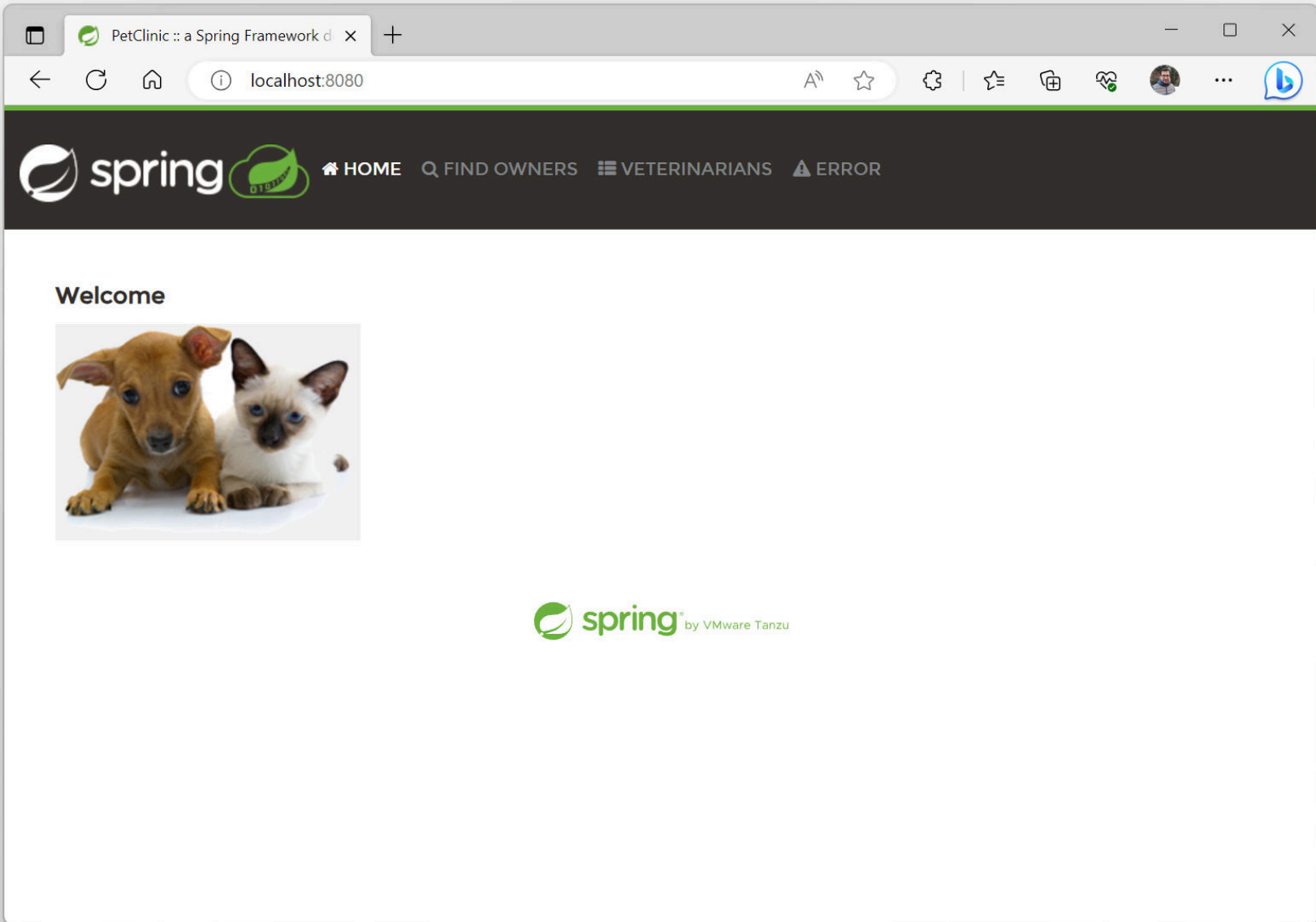
- How many vCPUs needs my application?
- How much RAM does it need?
- How much storage does it need?
- What technology stack should I use?
- What type of application do we build?

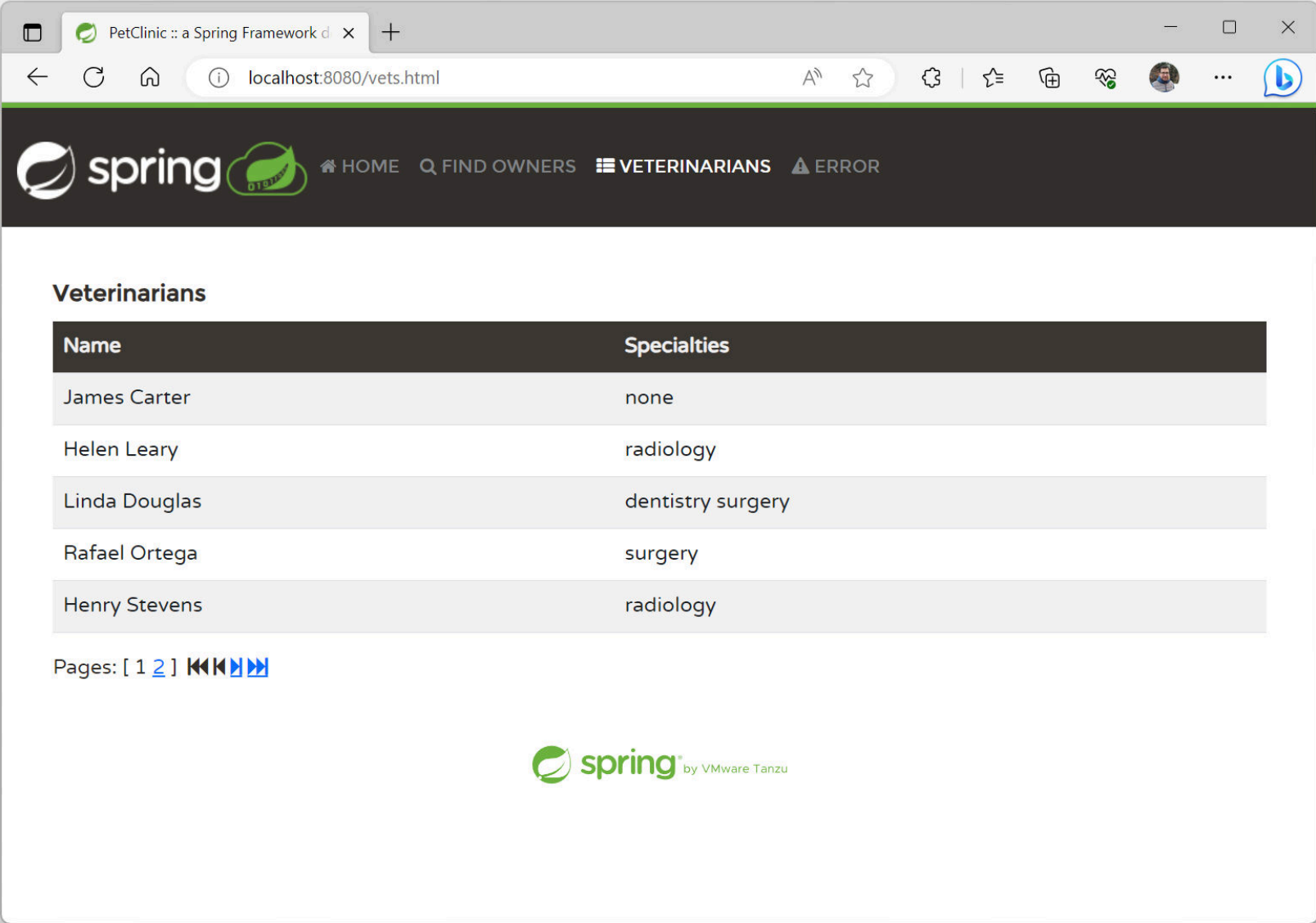


Agenda

Agenda

- Spring PetClinic & Baseline for comparison
- JVM Optimisations
- Spring Boot Optimisations
- Application Optimisations
- Other Runtimes
- Conclusions
- Some simple optimisations applied (OpenJDK examples)





Spring Petclinic Community

- spring-framework-petclinic
- spring-petclinic-angular(js)
- spring-petclinic-rest
- spring-petclinic-graphql
- spring-petclinic-microservices
- spring-petclinic-data-jdbc
- spring-petclinic-cloud
- spring-petclinic-mustache
- spring-petclinic-kotlin
- spring-petclinic-reactive
- spring-petclinic-hilla
- spring-petclinic-angularjs
- spring-petclinic-vaadin-flow
- spring-petclinic-reactjs
- spring-petclinic-htmx
- spring-petclinic-istio
- ...

Projects on GitHub: <https://github.com/spring-petclinic>

NO!

The official Spring PetClinic!  
Based on Spring Boot, Caffeine,
Thymeleaf, Spring Data JPA, H2 and
Spring MVC ...

Project on GitHub: <https://github.com/spring-projects/spring-petclinic>

Limitations

Limitations

- Docker stats prints wrong values for memory usage, use docker top instead. (See: <https://quarkus.io/guides/performance-measure#measuring-memory-correctly-on-docker>)

```
docker top <CONTAINER ID> -o pid,rss,args
```

- Thermal throttling of your notebook CPU may affect your results

Benchmark Steps

Build with Maven

Set the Java version and build the OCI container image.

```
sdk use java 17.0.18-librca
java -version

build_time=$(./mvnw clean spring-boot:build-image \
  | grep "Total time:" \
  | awk '{print $4 $5}')

echo -e "\033[0;32mBuild time: $build_time\033[0m"

image_size=$(docker images \
  | grep "spring-petclinic" \
  | awk '{print $7}')

echo -e "\033[0;32mImage Size: $image_size\033[0m"
```

Container Startup Time

Run the container with limited memory and measure the startup time.

```
docker container run -d --rm --memory="1g" -p 8080:8080 -t spring-petclinic:4.0.0-SNAPSHOT

# Get the container ID of the running container
container_id=$(docker ps -q)

startup_time=$(docker logs "$container_id"
| grep "Started" \
| awk '{print $14 $15}')

echo -e "\033[0;32mStartup time: $startup_time\033[0m"
```

Memory Usage after Startup

Extract the memory usage of the running container. Convert rss to MB.

```
memory=$(docker top $(docker ps -lq) -o pid,rss,args \
| tail -n +2 \
| awk '{ $2=int($2/1024)"MB"; }{ print $2; }')
echo -e "\033[0;32mStartup memory usage: $memory\033[0m"
```

Warm-up Application

Warm up the application with for 60s, json output and no UI.

```
oha --no-tui -z60s --disable-keepalive -j "http://localhost:8080/owners?page=2"
```

Load-Test Application

Load test during 60s, json output and no UI.

```
throughput=$(oha --no-tui -z60s --disable-keepalive -j "http://localhost:8080/owners?page=2" 2>&1 \  
| tee $filename \  
| jq '.summary.requestsPerSec')  
  
echo -e "\033[0;32mAvg Throughput: $throughput Requests/sec\033[0m"
```

Memory Usage after Load-Test

Extract the memory usage of the running container. Convert rss to MB.

```
memory=$(docker top $(docker ps -lq) -o pid,rss,args \  
| tail -n +2 \  
| awk '{ $2=int($2/1024)"MB"; } { print $2; }')  
  
echo -e "\033[0;32mEnd memory usage: $memory\033[0m"
```

Optimisation Experiments

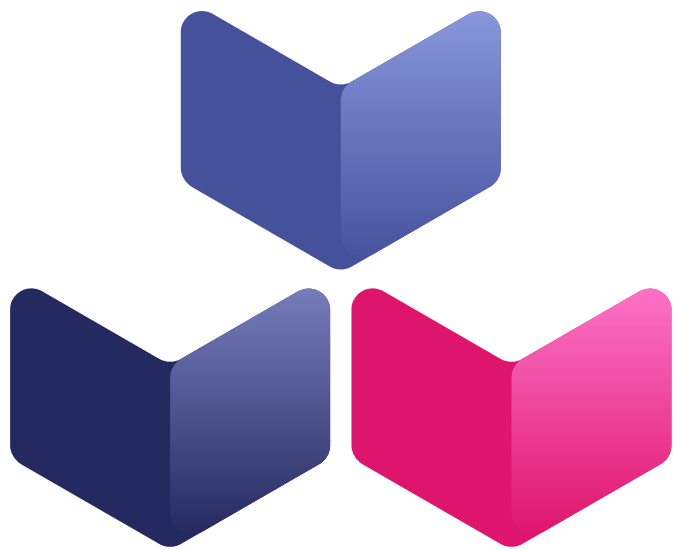
Baseline

Technology Stack

- OCI Container built with Buildpacks
- Java JDK 17 LTS
- Spring Boot 4.0
- Testcontainers
- DB migration using SQL scripts

Examination

- Build time
- Startup time
- Resource usage
- Container image size
- Throughput



Buildpacks.io



paketo

buildpacks



**Your app,
in your favorite language,
ready to run in the cloud**



GraalVM™



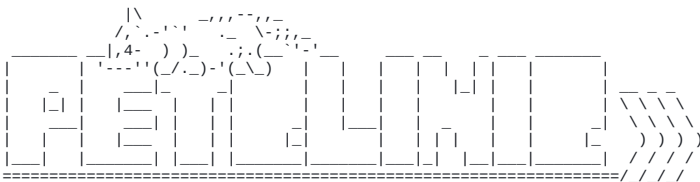
NGINX



```

Calculating JVM memory based on 22958184K available memory
For more information on this calculation, see https://paketo.io/docs/reference/java-reference/#memory-calculator
Calculated JVM Memory Configuration: -XX:MaxDirectMemorySize=10M -Xmx22316444K -XX:MaxMetaspaceSize=129739K -XX:ReservedCodeCacheSize=240M -Xss1M
(Total Memory: 22958184K, Thread Count: 250, Loaded Class Count: 20492, Headroom: 0%)
Enabling Java Native Memory Tracking
Adding 146 container CA certificates to JVM truststore
Spring Cloud Bindings Enabled
Picked up JAVA_TOOL_OPTIONS: -Djava.security.properties=/layers/paketo-buildpacks_bellsoft-liberica/java-security-properties/java-security.properties
-XX:+ExitOnOutOfMemoryError -XX:MaxDirectMemorySize=10M -Xmx22316444K -XX:MaxMetaspaceSize=129739K -XX:ReservedCodeCacheSize=240M -Xss1M
-XX:+UnlockDiagnosticVMOptions -XX:NativeMemoryTracking=summary -XX:+PrintNMTStatistics -Dorg.springframework.cloud.bindings.boot.enable=true

```



```
:: Built with Spring Boot :: 4.0.3
```

```

2026-03-07T18:00:07.524Z INFO 1 --- [main] o.s.s.petclinic.PetClinicApplication : Starting PetClinicApplication v4.0.0-SNAPSHOT using Java 17.0.18
with PID 1 (/workspace/BOOT-INF/classes started by cnb in /workspace)
2026-03-07T18:00:07.528Z INFO 1 --- [main] o.s.s.petclinic.PetClinicApplication : No active profile set, falling back to 1 default profile: "default"
2026-03-07T18:00:08.469Z INFO 1 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-03-07T18:00:08.527Z INFO 1 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 47 ms. Found 3 JPA repository interfaces.
2026-03-07T18:00:09.365Z INFO 1 --- [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-03-07T18:00:09.380Z INFO 1 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-03-07T18:00:09.381Z INFO 1 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.18]
2026-03-07T18:00:09.440Z INFO 1 --- [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1860 ms
2026-03-07T18:00:09.733Z INFO 1 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-03-07T18:00:09.937Z INFO 1 --- [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection conn0: url=jdbc:h2:mem:8a2f72dc-247f-4908-98aa-3a8d6d1fd45a user=SA
2026-03-07T18:00:09.940Z INFO 1 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-03-07T18:00:10.124Z INFO 1 --- [main] org.hibernate.orm.jpa : HHH000540: Processing PersistenceUnitInfo [name: default]
2026-03-07T18:00:10.182Z INFO 1 --- [main] org.hibernate.orm.core : HHH000001: Hibernate ORM core version 7.2.4.Final
2026-03-07T18:00:08.145Z INFO 1 --- [main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-03-07T18:00:08.205Z INFO 1 --- [main] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
Database JDBC URL [jdbc:h2:mem:8a2f72dc-247f-4908-98aa-3a8d6d1fd45a]
Database driver: H2 JDBC Driver
Database dialect: H2Dialect
Database version: 2.4.240
Default catalog/schema: 8A2F72DC-247F-4908-98AA-3A8D6D1FD45A/PUBLIC
Autocommit mode: undefined/unknown
Isolation level: READ_COMMITTED [default READ_COMMITTED]
JDBC fetch size: 100
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-03-07T18:00:09.147Z INFO 1 --- [main] org.hibernate.orm.core : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
2026-03-07T18:00:09.153Z INFO 1 --- [main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-03-07T18:00:09.250Z INFO 1 --- [main] o.s.d.j.r.query.QueryEnhancerFactories : Hibernate is in classpath; If applicable, HQL parser will be used.
2026-03-07T18:00:10.577Z INFO 1 --- [main] o.s.b.a.e.web.EndpointLinksResolver : Exposing 13 endpoints beneath base path '/actuator'
2026-03-07T18:00:10.688Z INFO 1 --- [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-03-07T18:00:10.701Z INFO 1 --- [main] o.s.s.petclinic.PetClinicApplication : Started PetClinicApplication in 3.621 seconds (process running for 6.498)

```

1000x better than your regular
Dockerfile  ...

... more **Secure**  and maintained by the
Buildpacks community.

See also: <https://buildpacks.io/> and <https://www.cncf.io/projects/buildpacks/>

Friends don't
let friends write
Dockerfiles!

@starbuxman



Benchmarks

Benchmarks

- Build
 - Maven build time
 - Artifact / Container Image size
- Startup
 - Startup time
 - Memory usage
- Throughput & Latency
 - `oha --no-tui -z60s --disable-keepalive <URL>`
 - 60s warmup, 60s measurement
 - Docker container with 2 vCPU and 1 GB RAM

No Optimizing - Baseline JDK 17

- Spring PetClinic (no adjustments)
- Bellsoft Liberica JDK 17.0.18
- Java Memory Calculator

```
sdk use java 17.0.18-librca
```

```
mvn spring-boot:build-image
```

```
docker run -p 8080:8080 -t spring-petclinic:4.0.0-SNAPSHOT
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms

No Optimizing - Baseline JDK 21

- Spring PetClinic (JDK 21 adjustments)
- Bellsoft Liberica JDK 21.0.10
- Java Memory Calculator

```
sdk use java 21.0.10-librca
```

```
mvn -Djava.version=21 spring-boot:build-image \  
    -Dspring-boot.build-image.imageName=spring-petclinic:4.0.0-SNAPSHOT-jdk21
```

```
docker run -p 8080:8080 -t spring-petclinic:4.0.0-SNAPSHOT-jdk21
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 21	65s	307MB	5.192s	275MB	246/s	373MB	190ms	296ms	410ms	706ms	1006ms

No Optimizing - Baseline JDK 25

- Spring PetClinic (JDK 25 adjustments)
- Bellsoft Liberica JDK 25.0.2
- Java Memory Calculator

```
sdk use java 25.0.2-librca
```

```
mvn -Djava.version=25 spring-boot:build-image \  
    -Dspring-boot.build-image.imageName=spring-petclinic:4.0.0-SNAPSHOT-jdk25
```

```
docker run -p 8080:8080 -t spring-petclinic:4.0.0-SNAPSHOT-jdk25
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 25	66s	375MB	5.124s	270MB	236/s	457MB	195ms	299ms	406ms	689ms	926ms

JVM Optimisations

-noverify

The verifier is turned off because some of the bytecode rewriting stretches the meaning of some bytecodes - in a way that doesn't bother the JVM, but does bother the verifier.

Warning: The `-Xverify:none` and `-noverify` options are deprecated in JDK 13 and are likely to be removed in a future release.

```
docker run -p 8080:8080 -e "JAVA_TOOL_OPTIONS=-noverify" \
-t spring-petclinic:4.0.0-SNAPSHOT
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	-	-	4.493s	247MB	215/s	366MB	200ms	311ms	493ms	804ms	1096ms
Java 21	-	-	4.367s	251MB	240/s	372MB	192ms	298ms	414ms	758ms	1014ms
Java 25	-	-	4.426s	249MB	238/s	431MB	193ms	299ms	409ms	694ms	915ms

-XX:+UseCompactObjectHeaders

The Compact Object Headers (JEP 384) feature reduces the memory footprint of Java objects by using a more compact representation for object headers. This can lead to improved cache performance and reduced memory usage.

```
docker run -p 8080:8080 -e "JAVA_TOOL_OPTIONS=-XX:+UseCompactObjectHeaders" \
-t spring-petclinic:4.0.0-SNAPSHOT
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
 Java 25	66s	375MB	5.124s	270MB	236/s	457MB	195ms	299ms	406ms	689ms	926ms
Java 25	-	-	5.004s	266MB	186/s	451MB	219ms	389ms	504ms	803ms	1037ms

-XX:TieredStopAtLevel=1

Tiered compilation is enabled by default (since Java 8). Unless explicitly specified, the JVM decides which JIT compiler to use based on our CPU. For multi-core processors or 64-bit VMs, the JVM will select C2.

To disable C2 and use only the C1 compiler with no profiling overhead, we can use the `-XX:TieredStopAtLevel=1` parameter.

```
docker run -p 8080:8080 -e "JAVA_TOOL_OPTIONS=-XX:TieredStopAtLevel=1" \
-t spring-petclinic:4.0.0-SNAPSHOT
```

It will slow down the JIT later at the expense of the saved startup time!

	Build	Image	Startup	Initial RAM	T...	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	-	-	4.118s	221MB	136/s	293MB	380ms	482ms	579ms	713ms	877ms
Java 21	-	-	4.237s	232MB	134/s	305MB	383ms	489ms	591ms	786ms	909ms
Java 25	-	-	3.933s	227MB	169/s	389MB	291ms	398ms	506ms	754ms	953ms

-XX:+UseParallelGC

The Parallel Garbage Collector (Parallel GC) is a high-throughput garbage collector that is designed to take advantage of multiple processors. It performs minor garbage collections in parallel, which can lead to shorter pause times and improved application performance.

```
docker run -p 8080:8080 -e "JAVA_TOOL_OPTIONS=-XX:+UseParallelGC" \
-t spring-petclinic:4.0.0-SNAPSHOT
```

	Build	Image	Startup	Initial RAM	T...	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	-	-	5.218s	318MB	238/s	464MB	194ms	299ms	414ms	713ms	1000ms
Java 21	-	-	5.015s	339MB	253/s	486MB	188ms	292ms	402ms	696ms	964ms
Java 25	-	-	4.924s	347MB	252/s	478MB	191ms	292ms	395ms	629ms	827ms

-XX:+UseZGC -XX:+ZGenerational

The Z Garbage Collector (ZGC) is a scalable, low-latency garbage collector. ZGC performs all expensive work concurrently without stopping application threads for more than 10ms, making it suitable for applications that require low latency and/or use a very large heap (multi-terabyte).

```
docker run -p 8080:8080 -e "JAVA_TOOL_OPTIONS=-XX:+UseZGC -XX:+ZGenerational" \
-t spring-petclinic:4.0.0-SNAPSHOT
```

	Build	Image	Startup	Initial RAM	T...	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 21	-	-	5.437s	527MB	228/s	685MB	195ms	303ms	478ms	785ms	1063ms
Java 25	-	-	5.436s	496MB	226/s	687MB	198ms	304ms	435ms	701ms	932ms

See also: <https://wiki.openjdk.org/display/zgc/Main>

VM Options Explorer

<https://chriswhocodes.com>

VM Options Explorer - OpenJDK11

<https://chriswhocodes.com>

Byte-Me FullJEP JEPMap JEPSearch hsdisk JITWatch JaCoLine VM Options Explorer VM Intrinsic Explorer GC Explorer Optimizing Java Thank You!

VM Options Explorer - OpenJDK11 HotSpot

[OpenJDK HotSpot](#)

[Options added/removed between JDKs](#)

OpenJDK options also hosted on [foojay.io](#)

6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

[Alibaba Dragonwell](#)

8	11	17	21

[Amazon Corretto](#)

8	11	17	19	20	21	22	24

[Azul Systems](#)

Platform Prime								Zulu									
8	11	13	15	17	19	8	11	13	15	16	17	18	19	20	21	22	24

[BellSoft Liberica](#)

8	11	17	18	19	20	21	22

[Eclipse Temurin](#)

8	11	17	18	19	20	21	22

[JDK-based GraalVM](#)

17		21		22		24	
JDK	Native	JDK	Native	JDK	Native	JDK	Native

[JetBrains Runtime](#)

11	17	21

[Microsoft](#)

11	16	17	21

[OpenJ9](#)

[OpenJ9](#)

[Oracle](#)

6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24

[SAP SapMachine](#)

11	17	19	20	21

Search OpenJDK11 HotSpot Options:

Name	Since	Deprecated	Type	OS	CPU	Component	Default	Availability	Description	Defined in
Show All	Show All	Show All	Show All	Show All	Show All	Show All	Show All	Show All	Show All	Show All

Spring Boot Optimisations

Lazy Spring Beans (1)

Configure lazy initialisation throughout your application. A Spring Boot property makes all beans lazy by default, initialising them only when needed. You can use `@Lazy` to override this behaviour, e.g. `@Lazy(false)`.

```
docker run -p 8080:8080 \
  -e spring.main.lazy-initialization=true \
  -e spring.data.jpa.repositories.bootstrap-mode=lazy \
  -t spring-petclinic:4.0.0-SNAPSHOT
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	-	-	4.575s	243MB	229/s	363MB	196ms	302ms	457ms	782ms	1057ms

Lazy Spring Beans (2)

Pros

- Faster startup useful in cloud environments
- Application startup is a CPU-intensive task. Spreads load over time

Cons

- Initial requests may take longer
- Class loading problems and misconfigurations not detected at startup
- Beans creation errors only be found when the bean is loaded

Fixing Spring Boot Config Location

Determine the location of the Spring Boot configuration file(s).
Considered in the following order (`application.properties` and YAML variants)

- Application properties packaged in your jar
- Profile-specific application properties packaged within your jar
- Application properties outside your packaged jar
- Profile-specific application properties outside your packaged jar

```
docker run -p 8080:8080 -e spring.config.location=classpath:application.properties \
  t.spring.notelinic:4.0.0-SNAPSHOT
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	-	-	5.476s	253MB	225/s	364MB	198ms	304ms	480ms	788ms	1061ms

No Spring Boot Actuators

Don't use actuators if you can afford not to 😊.

- Number of Spring Beans
 - Spring Pet Clinic with actuators: 457
 - Spring Pet Clinic without actuators: 282 🔥

```
sdk use java 17.0.18-librca
```

```
mvn spring-boot:build-image
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
🕒 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	67s	277MB	4.636s	254MB	240/s	361MB	193ms	299ms	415ms	715ms	1005ms
Java 21	63s	305MB	4.469s	254MB	264/s	367MB	183ms	287ms	399ms	693ms	956ms
Java 25	64s	373MB	4.596s	253MB	256/s	468MB	187ms	290ms	398ms	639ms	890ms

Ahead-of-Time Processing (AOT) (1)

Spring AOT is a process that analyses your application at build time and generates an optimised version of it.

As the `BeanFactory` is fully prepared at build time, conditions are also evaluated. E.g. in `Configurations`, `Component- & Entity-Scan`, `@Profile`, `@Conditional`, etc.

Spring AOT serves as a replacement for `spring-context-indexer` since Spring Framework 6.1 and Spring Boot 3.2.

Ahead-of-Time Processing (AOT) (2)

```
<plugin>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-maven-plugin</artifactId>
  <configuration>
    <image>
      <env>
        <BP_SPRING_AOT_ENABLED>true</BP_SPRING_AOT_ENABLED>
      </env>
    </image>
  </configuration>
  <executions>
    <execution>
      <id>process-aot</id>
      <goals>
        <goal>process-aot</goal>
      </goals>
    </execution>
  </executions>
</plugin>
```

Ahead-of-Time Processing (AOT) (3)

We create a new container image with the AOT-processed application. The Buildpack enables the property `spring.aot.enabled=true`

```
sdk use java 17.0.18-librca  
  
mvn spring-boot:build-image  
  
docker run -p 8080:8080 -e spring.aot.enabled=true \  
-t spring-petclinic-aot:4.0.0-SNAPSHOT
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	73s	281MB	4.249s	241MB	227/s	354MB	197ms	303ms	460ms	756ms	1071ms
Java 21	75s	308MB	4.078s	241MB	253/s	364MB	187ms	292ms	403ms	697ms	961ms
Java 25	75s	377MB	4.141s	238MB	228/s	554MB	197ms	303ms	414ms	694ms	931ms

Disabling JMX

JMX is `spring.jmx.enabled=false` by default in Spring Boot since 2.2.0 and later. Setting `BPL_JMX_ENABLED=true` and `BPL_JMX_PORT=9999` on the container will add the following arguments to the `java` command.

```
-Djava.rmi.server.hostname=127.0.0.1  
-Dcom.sun.management.jmxremote.authenticate=false  
-Dcom.sun.management.jmxremote.ssl=false  
-Dcom.sun.management.jmxremote.port=9999  
-Dcom.sun.management.jmxremote.rmi.port=9999
```

```
docker run -p 8080:8080 -p 9999:9999 \  
  -e BPL_JMX_ENABLED=false \  
  -e BPL_JMX_PORT=9999 \  
  -e spring.jmx.enabled=false \  
  -t spring-petclinic:4.0.0-SNAPSHOT
```

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Application Optimisations

Dependency Cleanup (2)

DepClean detects all unused dependencies declared in the `pom.xml` file of a project and creates a `pom-debloated.xml`. The generated report shows possible unused dependencies.

```
<plugin>
  <groupId>se.kth.castor</groupId>
  <artifactId>depclean-maven-plugin</artifactId>
  <version>2.2.0-SNAPSHOT</version>
  <executions>
    <execution>
      <goals>
        <goal>depclean</goal>
      </goals>
    </execution>
  </executions>
</plugin>
```

```
mvn se.kth.castor:depclean-maven-plugin:2.2.0-SNAPSHOT:depclean -DfailIfUnusedDirect=true -DignoreScopes=provided,test,runtime,system,import
```

```

-----
D E P C L E A N   A N A L Y S I S   R E S U L T S
-----
USED DIRECT DEPENDENCIES [10]:
  com.h2database:h2:2.4.240:runtime (2 MB)
  com.mysql:mysql-connector-j:9.6.0:runtime (2 MB)
  org.postgresql:postgresql:42.7.10:runtime (1 MB)
  com.github.ben-manes.caffeine:caffeine:3.2.3:runtime (996 KB)
  jakarta.xml.bind:jakarta.xml.bind-api:4.0.4:compile (128 KB)
  org.springframework.boot:spring-boot-testcontainers:4.0.3:test (74 KB)
  ...
USED TRANSITIVE DEPENDENCIES [107]:
  org.testcontainers:testcontainers:2.0.3:test (16 MB)
  org.hibernate.orm:hibernate-core:7.2.4.Final:compile (13 MB)
  net.bytebuddy:byte-buddy:1.17.8:runtime (8 MB)
  org.apache.tomcat.embed:tomcat-embed-core:11.0.18:compile (3 MB)
  com.github.docker-java:docker-java-transport-zero-dep:3.7.0:test (2 MB)
  org.springframework:spring-web:7.0.5:compile (2 MB)
  ...
USED INHERITED DIRECT DEPENDENCIES [0]:
USED INHERITED TRANSITIVE DEPENDENCIES [0]:
POTENTIALLY UNUSED DIRECT DEPENDENCIES [14]:
  org.webjars.npm:bootstrap:5.3.8:runtime (1 MB)
  org.webjars.npm:font-awesome:4.7.0:runtime (665 KB)
  org.springframework.boot:spring-boot-devtools:4.0.3:compile (208 KB)
  org.springframework.boot:spring-boot-docker-compose:4.0.3:test (123 KB)
  org.springframework.boot:spring-boot-starter-actuator:4.0.3:compile (4 KB)
  ...
POTENTIALLY UNUSED TRANSITIVE DEPENDENCIES [34]:
  commons-codec:commons-codec:1.19.0:test (365 KB)
  org.springframework.boot:spring-boot-actuator:4.0.3:compile (348 KB)
  org.yaml:snakeyaml:2.5:compile (332 KB)
  org.attoparser:attoparser:2.0.7.RELEASE:compile (240 KB)
  org.springframework.boot:spring-boot-micrometer-metrics:4.0.3:compile (235 KB)
  org.thymeleaf:thymeleaf-spring6:3.1.3.RELEASE:compile (184 KB)
  org.hdrhistogram:HdrHistogram:2.2.2:runtime (173 KB)
  org.unbescape:unbescape:1.1.6.RELEASE:compile (169 KB)
  org.springframework.boot:spring-boot-web-server:4.0.3:compile (165 KB)
  org.springframework.boot:spring-boot-health:4.0.3:compile (150 KB)
  ...
POTENTIALLY UNUSED INHERITED DIRECT DEPENDENCIES [0]:
POTENTIALLY UNUSED INHERITED TRANSITIVE DEPENDENCIES [0]:
[INFO] Analysis done in 0min 27s

```

Dependency Cleanup (2)

But there are some challenges:

- Component & Entity Scanning through Classpath Scanning
- Spring Boot uses `META-INF/spring-boot/org.springframework.boot.autoconfigure.AutoConfiguration.imports`
- Spring XML Configuration and `web.xml`

```
sdk use java 17.0.18-librca
```

```
mvn spring-boot:build-image
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	37s	270MB	4.372s	238MB	247/s	349MB	185ms	296ms	413ms	736ms	1039ms
Java 21	37s	297MB	4.17s	244MB	275/s	357MB	171ms	283ms	395ms	691ms	979ms
Java 25	37s	366MB	4.276s	243MB	263/s	458MB	183ms	287ms	397ms	639ms	894ms

More Application Optimisations

- Application Optimisations
- Better Connection Pooling
- Improved Data Caching
- Database Queries Optimisations
- Batch Processing Optimisations
- Reduce Garbage Collection
- Monitor and Tune JVM Settings
- Reduce Overall Memory Footprint
- Async Logging

Even More JVM Optimisations

JLink (1)

`jlink` assembles and optimises a set of modules and their dependencies into a custom runtime image for your application.

```
$ jlink \  
  --add-modules java.base, ... \  
  --strip-debug \  
  --no-man-pages \  
  --no-header-files \  
  --compress=2 \  
  --output /javaruntime
```

```
$ /javaruntime/bin/java HelloWorld  
Hello, World!
```

JLink (2)

```
<plugin>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-maven-plugin</artifactId>
  <configuration>
    <image>
      <env>
        <BP_JVM_JLINK_ENABLED>true</BP_JVM_JLINK_ENABLED>
      </env>
    </image>
  </configuration>
</plugin>
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	74s	190MB	5.334s	265MB	227/s	375MB	197ms	303ms	474ms	780ms	1017ms
Java 21	77s	198MB	5.117s	285MB	245/s	390MB	191ms	296ms	408ms	701ms	996ms
Java 25	72s	201MB	5.057s	281MB	260/s	387MB	186ms	289ms	394ms	611ms	870ms

Java Dependency Analysis Tool

`jdeps` is a Java command-line tool that analyzes class and jar files to list the package-level or class-level dependencies, helping developers understand module dependencies and improve code modularity.

```
jar xvf target/spring-petclinic-4.0.0-SNAPSHOT.jar

jdeps --ignore-missing-deps -q \
      --recursive \
      --multi-release 17 \
      --print-module-deps \
      --class-path 'BOOT-INF/lib/*' \
      target/spring-petclinic-4.0.0-SNAPSHOT.jar
```

```
java.base,java.compiler,java.desktop,java.instrument,java.net.http,java.prefs,java.rmi,java.scripting,java.security.jgss,java.sql.rowset,jdk.jfr,jdk.management,jdk.unsupported
```

JLink (3)

```
<plugin>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-maven-plugin</artifactId>
  <configuration>
    <image>
      <env>
        <BP_JVM_JLINK_ENABLED>true</BP_JVM_JLINK_ENABLED>
        <BP_JVM_JLINK_ARGS>--add-modules java.base,java.compiler,java.desktop,java.instrument,
        java.net.http,java.prefs,java.rmi,java.scripting,java.security.jgss,java.sql.rowset,
        jdk.jfr,jdk.management,jdk.unsupported --compress=2 --no-header-files --no-man-pages
        --strip-debug</BP_JVM_JLINK_ARGS>
      </env>
    </image>
  </configuration>
</plugin>
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	71s	179MB	5.283s	267MB	227/s	373MB	196ms	303ms	464ms	786ms	1018ms
Java 21	72s	187MB	5.039s	284MB	248/s	387MB	188ms	294ms	407ms	704ms	990ms
Java 25	70s	189MB	5.124s	275MB	263/s	379MB	185ms	287ms	391ms	607ms	855ms

App CDS (1)

Class Data Sharing (CDS) is a JVM feature that can help reduce the startup time and memory footprint of Java applications. It allows classes to be pre-processed into a shared archive file that can be memory-mapped at runtime.

```
<plugin>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-maven-plugin</artifactId>
  <configuration>
    <image>
      <env>
        <BP_JVM_CDS_ENABLED>true</BP_JVM_CDS_ENABLED>
      </env>
    </image>
  </configuration>
</plugin>
```

App CDS (2)

To create the archive, two additional JVM flags must be specified:

- `-XX:ArchiveClassesAtExit=application.jsa` : creates the CDS archive on exit
- `-Dspring.context.exit=onRefresh` : starts and then immediately exits

Once the archive is available, the JVM can be started with the additional flag:

- `-XX:SharedArchiveFile=application.jsa` : to use it

```
sdk use java 17.0.18-librca
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	80s	450MB	2.767s	239MB	258/s	353MB	183ms	291ms	403ms	697ms	938ms
Java 21	79s	476MB	2.72s	247MB	279/s	366MB	143ms	280ms	394ms	680ms	920ms

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AOT Caching

Project Leyden is an OpenJDK project that aims to improve the startup time and reduce the memory footprint of Java applications by introducing AOT (Ahead-of-Time) compilation and class data sharing (CDS) enhancements.

Training the AOT cache requires two additional JVM flags:

`-XX:AOTCacheOutput=app.aot -Dspring.context.exit=onRefresh`

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 25	87s	570MB	2.078s	235MB	271/s	417MB	182ms	284ms	388ms	600ms	803ms

Other Runtimes



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Faster Startup

Faster Ramp-up, when deployed to cloud

Smaller

Our Story

Eclipse OpenJ9

```
<plugin>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-maven-plugin</artifactId>
  <configuration>
    <image>
      <buildpacks>
        <buildpack>paketobuildpacks/eclipse-openj9:latest</buildpack>
        <!-- Used to inherit all the other Java buildpacks -->
        <buildpack>urn:cnb:builder:paketo-buildpacks/java</buildpack>
      </buildpacks>
    </image>
  </configuration>
</plugin>
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	75s	266MB	5.723s	189MB	82/s	277MB	606ms	773ms	913ms	1302ms	1623ms
Java 21	74s	284MB	5.616s	184MB	80/s	275MB	611ms	788ms	983ms	1311ms	1601ms
74s	283MB	6.155s	186MB	70/s	270MB	712ms	885ms	1007ms	1400ms	1657ms	

GraalVM CE

```
<plugin>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-maven-plugin</artifactId>
  <configuration>
    <image>
      <buildpacks>
        <buildpack>paketobuildpacks/graalvm:latest</buildpack>
        <!-- Used to inherit all the other Java buildpacks -->
        <buildpack>urn:cnb:builder:paketo-buildpacks/java</buildpack>
      </buildpacks>
    </image>
  </configuration>
</plugin>
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	83s	677MB	5.423s	290MB	179/s	388MB	220ms	398ms	592ms	994ms	1359ms
Java 21	83s	663MB	5.307s	312MB	188/s	408MB	208ms	375ms	565ms	965ms	1327ms
Java 25	84s	851MB	5.447s	323MB	211/s	419MB	200ms	318ms	496ms	801ms	1088ms

Oracle JDK

```
<plugin>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-maven-plugin</artifactId>
  <configuration>
    <image>
      <buildpacks>
        <buildpack>paketobuildpacks/oracle:latest</buildpack>
        <!-- Used to inherit all the other Java buildpacks -->
        <buildpack>urn:cnb:builder:paketo-buildpacks/java</buildpack>
      </buildpacks>
    </image>
  </configuration>
</plugin>
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 21	73s	448MB	5.181s	274MB	241/s	389MB	192ms	298ms	415ms	712ms	995ms
Java 25	73s	492MB	5.046s	274MB	242/s	458MB	192ms	297ms	404ms	668ms	894ms

Azul Zulu

```
<plugin>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-maven-plugin</artifactId>
  <configuration>
    <image>
      <buildpacks>
        <buildpack>paketobuildpacks/azul-zulu:latest</buildpack>
        <!-- Used to inherit all the other Java buildpacks -->
        <buildpack>urn:cnb:builder:paketo-buildpacks/java</buildpack>
      </buildpacks>
    </image>
  </configuration>
</plugin>
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	67s	243MB	5.298s	256MB	224/s	368MB	198ms	306ms	481ms	793ms	1093ms
Java 21	66s	267MB	5.745s	276MB	247/s	384MB	190ms	295ms	407ms	700ms	995ms
Java 25	65s	301MB	5.131s	273MB	250/s	390MB	190ms	294ms	398ms	630ms	862ms

Other Buildpack Builders

Bellsoft Buildpack Builder

Bellsoft provides an optimised builder for Spring Boot applications. It uses the Bellsoft Alpaquita, Liberica JDK and the musl C library. A glibc version is also available.

```
<image>  
  <builder>bellsoft/buildpacks.builder:musl</builder>  
  <!--builder>bellsoft/buildpacks.builder:glibc</builder-->  
</image>
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17 (m)	65s	182MB	5.094s	280MB	232/s	384MB	195ms	301ms	443ms	763ms	1038ms
Java 21 (m)	61s	191MB	4.905s	298MB	261/s	400MB	184ms	289ms	399ms	689ms	909ms
Java 25 (m)	60s	197MB	5.013s	295MB	270/s	403MB	185ms	282ms	382ms	592ms	788ms
Java 17 (g)	64s	189MB	5.137s	268MB	234/s	380MB	194ms	300ms	419ms	715ms	1009ms
Java 21 (g)	65s	197MB	4.93s	288MB	261/s	394MB	184ms	288ms	400ms	688ms	936ms
Java 25 (g)	64s	203MB	4.938s	282MB	251/s	383MB	190ms	293ms	397ms	631ms	839ms

Buildpack Jammy Builder Tiny

It's based on Ubuntu Jammy, and works with current JAVA versions. While the base image (default) is bigger the tiny is intended to be used with GraalVM native images.

```
<plugin>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-maven-plugin</artifactId>
  <configuration>
    <image>
      <builder>paketobuildpacks/builder-jammy-java-tiny</builder>
    </image>
  </configuration>
</plugin>
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 17	66s	280MB	5.513s	259MB	215/s	364MB	201ms	310ms	491ms	800ms	1089ms
Java 21	66s	307MB	5.302s	274MB	235/s	379MB	194ms	300ms	418ms	712ms	995ms
Java 25	65s	376MB	5.129s	268MB	225/s	455MB	198ms	305ms	419ms	701ms	958ms

GraalVM Native Image (CE & Oracle)

A native image is a technology for building Java code into a standalone executable. This executable contains the application classes, classes from its dependencies, runtime library classes and statically linked native code from the JDK. The JVM is packaged in the native image, so there's no need for a Java Runtime Environment on the target system, but the build artifact is platform dependent.

```
mvn -Pnative spring-boot:build-image
```

```
docker run -p 8080:8080 -t spring-petclinic-native:4.0.0-SNAPSHOT
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
CE 25	242s	233MB	0.247s	205MB	1008/s	365MB	48ms	73ms	89ms	117ms	152ms
OG 25	345s	230MB	0.183s	176MB	1191/s	276MB	35ms	62ms	78ms	105ms	144ms

CRaC - OpenJDK (1)

CRaC (Coordinated Restore at Checkpoint) is a feature that allows you to take a snapshot of the state of a Java application and restart it from that state.

Currently only available from:

- Azul Zulu
- Bellsoft Liberica

The application starts within milliseconds!

.. with many other challenges like: embedded configs and secrets, same OS (Linux) and CPU Architecture, same JVM versions, and more ...

CRaC - OpenJDK (2)

```
export JAVA_HOME=/opt/openjdk-17-crac+6_linux-x64/  
export PATH=$JAVA_HOME/bin:$PATH
```

```
mvn clean verify
```

```
java -XX:CRaCCheckpointTo=crac-files -jar target/spring-petclinic-crac-4.0.0-SNAPSHOT.jar
```

```
jcmd target/spring-petclinic-crac-4.0.0-SNAPSHOT.jar JDK.checkpoint
```

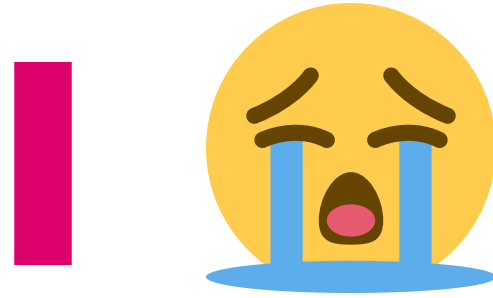
```
java -XX:CRaCRestoreFrom=crac-files
```

CRaC - OpenJDK (3)

CRaC is currently in an experimental state and has the following limitations.

- Since Spring Boot 3.2 CraC support finalised
 - Spring Framework 6.1.0
- Only available for Linux x86 / ARM 64 Bit
- Azul Zulu Build of OpenJDK with CRaC support for development purposes
 - Available for Windows and macOS
 - Simulated checkpoint/restore mechanism for development and testing

Other JVM vendors have similar features, e.g. OpenJ9 with CRIU support.



**No CRaC Buildpacks
for Java & Spring Boot 🍃**

Virtual Threads

A thread is the smallest unit of processing that can be scheduled. It runs concurrently with - and largely independently of - other such units. It is an instance of `java.lang.Thread`. There are two types of threads, platform threads and virtual threads.

```
sdk use java 21.0.10-librca
```

```
mvn spring-boot:build-image
```

```
docker run -e spring.threads.virtual.enabled=true \
```

	Build	Image	Startup	Initial RAM	Requests/s	RAM	50%	75%	90%	99%	99.9%
 Java 17	69s	279MB	5.509s	258MB	220/s	366MB	199ms	307ms	485ms	790ms	1059ms
Java 21 	-	-	5.535s	272MB	282/s	369MB	190ms	198ms	204ms	299ms	1367ms
Java 24 	-	-	5.223s	271MB	1571/s	368MB	29ms	32ms	35ms	110ms	779ms
Java 25 	-	-	5.077s	268MB	1538/s	376MB	28ms	33ms	35ms	110ms	983ms

Virtual Threads Java 21

CPU	VT	Startup	Initial RAM	Request/s	RAM	50%	75%	90%	99%	99.9%	99.99%
4 vCPU		4.634s	294MB	1559/s	408MB	28ms	46ms	62ms	89ms	115ms	151ms
4 vCPU		4.409s	295MB	1720/s	402MB	25ms	38ms	43ms	50ms	216ms	469ms
2 vCPU		5.075s	275MB	235/s	378MB	194ms	301ms	419ms	715ms	1006ms	1222ms
2 vCPU		5.535s	272MB	282/s	369MB	190ms	198ms	204ms	299ms	1367ms	2522ms
1 vCPU		14.289s	275MB	76/s	364MB	573ms	924ms	1372ms	2347ms	3198ms	3670ms
1 vCPU		13.828s	271MB	99/s	360MB	499ms	529ms	598ms	807ms	2471ms	3548ms
0.5 vCPU		41.001s	271MB	22/s	349MB	2024ms	3077ms	4293ms	6614ms	8599ms	9286ms
0.5 vCPU		41.228s	272MB	32/s	354MB	1500ms	1701ms	1899ms	2695ms	7450ms	9776ms
0.25 vCPU		87.302s	271MB	10/s	343MB	4625ms	7350ms	10304ms	16475ms	21732ms	21732ms
0.25 vCPU		87.500s	271MB	11/s	340MB	4501ms	5072ms	5807ms	9811ms	14652ms	14652ms

Virtual Threads Java 24

CPU	VT	Startup	Initial RAM	Request/s	RAM	50%	75%	90%	99%	99.9%	99.99%
4 vCPU		4.143s	288MB	1544/s	419MB	31ms	43ms	55ms	74ms	92ms	113ms
4 vCPU		4.174s	288MB	2140/s	669MB	21ms	33ms	44ms	69ms	92ms	117ms
2 vCPU		5.261s	270MB	277/s	386MB	180ms	272ms	379ms	600ms	802ms	1044ms
2 vCPU		5.223s	271MB	1571/s	368MB	29ms	32ms	35ms	110ms	779ms	1390ms
1 vCPU		14.561s	268MB	81/s	370MB	504ms	874ms	1274ms	2151ms	3025ms	3928ms
1 vCPU		15.066s	267MB	706/s	358MB	60ms	71ms	112ms	179ms	200ms	221ms
0.5 vCPU		43.477s	267MB	24/s	350MB	1829ms	2950ms	4199ms	6565ms	8690ms	9548ms
0.5 vCPU		42.973s	263MB	57/s	348MB	805ms	1000ms	1198ms	1751ms	2778ms	3904ms
0.25 vCPU		91.801s	265MB	10/s	327MB	4427ms	7076ms	9849ms	15690ms	20250ms	20250ms
0.25 vCPU		91.747s	265MB	14/s	321MB	3526ms	4073ms	4522ms	8043ms	11059ms	11059ms

Virtual Threads Java 25

CPU	VT	Startup	Initial RAM	Request/s	RAM	50%	75%	90%	99%	99.9%	99.99%
4 vCPU		4.011s	295MB	1720/s	402MB	25ms	38ms	43ms	50ms	216ms	469ms
4 vCPU		4.018s	286MB	2099/s	685MB	19ms	36ms	54ms	91ms	127ms	161ms
2 vCPU		5.535s	272MB	282/s	369MB	190ms	198ms	204ms	299ms	1367ms	2522ms
2 vCPU		5.077s	268MB	1538/s	376MB	28ms	33ms	35ms	110ms	983ms	1893ms
1 vCPU		13.828s	271MB	99/s	360MB	499ms	529ms	598ms	807ms	2471ms	3548ms
1 vCPU		13.944s	267MB	701/s	363MB	60ms	75ms	114ms	181ms	216ms	274ms
0.5 vCPU		41.228s	272MB	32/s	354MB	1500ms	1701ms	1899ms	2695ms	7450ms	9776ms
0.5 vCPU		42.443s	264MB	59/s	352MB	774ms	922ms	1099ms	1720ms	5925ms	7505ms
0.25 vCPU		87.5s	271MB	11/s	340MB	4501ms	5072ms	5807ms	9811ms	14652ms	14652ms
0.25 vCPU		88.377s	263MB	15/s	322MB	3102ms	3628ms	4129ms	10626ms	21994ms	21994ms

Summary

Summary

- No Optimisations with JDK 17 & JDK 21, ...
- JVM Tuning
- Lazy Spring Beans
- No Spring Boot Actuators
- Fix Spring Boot Config Location
- Disabling JMX
- Dependency Cleanup
- Ahead-of-Time Processing (AOT)
- JLink
- Other JVMs (Eclipse OpenJ9, GraalVM, OpenJDK with CRaC)
- GraalVM Native Image

Conclusions

Conclusions (1)

CPUs

- Your application may not need a full CPU at runtime.
- It will need several CPUs to start as fast as possible (at least 2, 4 is better).
- If you don't mind a slower startup, you can throttle the CPUs below 4.

See: <https://spring.io/blog/2018/11/08/spring-boot-in-a-container>

Conclusions (2)

Request/s

- Every application is different and has different requirements.
- Proper load testing can help find the optimal configuration for your application.

Conclusions (3)

Other Runtimes

- CRIU Support for OpenJDK and OpenJ9 is promising.
 - Supported by Spring since Spring Boot 3.2 / Spring Framework 6.1
- GraalVM Native Image is a great option for Java applications
 - But build times are long
 - The result is different from what you run in your IDE
- Eclipse OpenJ9 is an excellent option for running applications with less memory
 - But startup times are longer than with HotSpot.
- Depending on the distribution, you may get other exciting features.
 - Oracle GraalVM Enterprise Edition, Azul Platform Prime, IBM Semeru Runtime, ...

Conclusions (4)

Other Ideas

- CRaC (Coordinated Restore at Checkpoint)*
- Tree shaking / Obfuscator such as ProGuard*
- AutoConfiguration Filter -> **spring-boot-autoconfiguration-optimizer**
- More JVM tuning (GC, Memory, etc.)
- Project Leyden

A Few Simple Optimisations Applied

```
1  def fib(n):  
2      if n < 2:  
3          return n  
4      return fib(n-1) + fib(n-2)  
5  
6  def fib2(n):  
7      a, b = 0, 1  
8      for i in range(n):  
9          a, b = b, a + b  
10     return a  
11  
12  def fib3(n):  
13     if n < 2:  
14         return n  
15     a, b = 0, 1  
16     for i in range(n-2):  
17         a, b = b, a + b  
18     return b  
19  
20  def fib4(n):  
21     if n < 2:  
22         return n  
23     a, b = 0, 1  
24     for i in range(n-2):  
25         a, b = b, a + b  
26     return b  
27  
28  def fib5(n):  
29     if n < 2:  
30         return n  
31     a, b = 0, 1  
32     for i in range(n-2):  
33         a, b = b, a + b  
34     return b  
35  
36  def fib6(n):  
37     if n < 2:  
38         return n  
39     a, b = 0, 1  
40     for i in range(n-2):  
41         a, b = b, a + b  
42     return b  
43  
44  def fib7(n):  
45     if n < 2:  
46         return n  
47     a, b = 0, 1  
48     for i in range(n-2):  
49         a, b = b, a + b  
50     return b  
51  
52  def fib8(n):  
53     if n < 2:  
54         return n  
55     a, b = 0, 1  
56     for i in range(n-2):  
57         a, b = b, a + b  
58     return b  
59  
60  def fib9(n):  
61     if n < 2:  
62         return n  
63     a, b = 0, 1  
64     for i in range(n-2):  
65         a, b = b, a + b  
66     return b  
67  
68  def fib10(n):  
69     if n < 2:  
70         return n  
71     a, b = 0, 1  
72     for i in range(n-2):  
73         a, b = b, a + b  
74     return b  
75  
76  def fib11(n):  
77     if n < 2:  
78         return n  
79     a, b = 0, 1  
80     for i in range(n-2):  
81         a, b = b, a + b  
82     return b  
83  
84  def fib12(n):  
85     if n < 2:  
86         return n  
87     a, b = 0, 1  
88     for i in range(n-2):  
89         a, b = b, a + b  
90     return b  
91  
92  def fib13(n):  
93     if n < 2:  
94         return n  
95     a, b = 0, 1  
96     for i in range(n-2):  
97         a, b = b, a + b  
98     return b  
99  
100 def fib14(n):  
101     if n < 2:  
102         return n  
103     a, b = 0, 1  
104     for i in range(n-2):  
105         a, b = b, a + b  
106     return b  
107  
108 def fib15(n):  
109     if n < 2:  
110         return n  
111     a, b = 0, 1  
112     for i in range(n-2):  
113         a, b = b, a + b  
114     return b  
115  
116 def fib16(n):  
117     if n < 2:  
118         return n  
119     a, b = 0, 1  
120     for i in range(n-2):  
121         a, b = b, a + b  
122     return b  
123  
124 def fib17(n):  
125     if n < 2:  
126         return n  
127     a, b = 0, 1  
128     for i in range(n-2):  
129         a, b = b, a + b  
130     return b  
131  
132 def fib18(n):  
133     if n < 2:  
134         return n  
135     a, b = 0, 1  
136     for i in range(n-2):  
137         a, b = b, a + b  
138     return b  
139  
140 def fib19(n):  
141     if n < 2:  
142         return n  
143     a, b = 0, 1  
144     for i in range(n-2):  
145         a, b = b, a + b  
146     return b  
147  
148 def fib20(n):  
149     if n < 2:  
150         return n  
151     a, b = 0, 1  
152     for i in range(n-2):  
153         a, b = b, a + b  
154     return b  
155  
156 def fib21(n):  
157     if n < 2:  
158         return n  
159     a, b = 0, 1  
160     for i in range(n-2):  
161         a, b = b, a + b  
162     return b  
163  
164 def fib22(n):  
165     if n < 2:  
166         return n  
167     a, b = 0, 1  
168     for i in range(n-2):  
169         a, b = b, a + b  
170     return b  
171  
172 def fib23(n):  
173     if n < 2:  
174         return n  
175     a, b = 0, 1  
176     for i in range(n-2):  
177         a, b = b, a + b  
178     return b  
179  
180 def fib24(n):  
181     if n < 2:  
182         return n  
183     a, b = 0, 1  
184     for i in range(n-2):  
185         a, b = b, a + b  
186     return b  
187  
188 def fib25(n):  
189     if n < 2:  
190         return n  
191     a, b = 0, 1  
192     for i in range(n-2):  
193         a, b = b, a + b  
194     return b  
195  
196 def fib26(n):  
197     if n < 2:  
198         return n  
199     a, b = 0, 1  
200     for i in range(n-2):  
201         a, b = b, a + b  
202     return b  
203  
204 def fib27(n):  
205     if n < 2:  
206         return n  
207     a, b = 0, 1  
208     for i in range(n-2):  
209         a, b = b, a + b  
210     return b  
211  
212 def fib28(n):  
213     if n < 2:  
214         return n  
215     a, b = 0, 1  
216     for i in range(n-2):  
217         a, b = b, a + b  
218     return b  
219  
220 def fib29(n):  
221     if n < 2:  
222         return n  
223     a, b = 0, 1  
224     for i in range(n-2):  
225         a, b = b, a + b  
226     return b  
227  
228 def fib30(n):  
229     if n < 2:  
230         return n  
231     a, b = 0, 1  
232     for i in range(n-2):  
233         a, b = b, a + b  
234     return b  
235  
236 def fib31(n):  
237     if n < 2:  
238         return n  
239     a, b = 0, 1  
240     for i in range(n-2):  
241         a, b = b, a + b  
242     return b  
243  
244 def fib32(n):  
245     if n < 2:  
246         return n  
247     a, b = 0, 1  
248     for i in range(n-2):  
249         a, b = b, a + b  
250     return b  
251  
252 def fib33(n):  
253     if n < 2:  
254         return n  
255     a, b = 0, 1  
256     for i in range(n-2):  
257         a, b = b, a + b  
258     return b  
259  
260 def fib34(n):  
261     if n < 2:  
262         return n  
263     a, b = 0, 1  
264     for i in range(n-2):  
265         a, b = b, a + b  
266     return b  
267  
268 def fib35(n):  
269     if n < 2:  
270         return n  
271     a, b = 0, 1  
272     for i in range(n-2):  
273         a, b = b, a + b  
274     return b  
275  
276 def fib36(n):  
277     if n < 2:  
278         return n  
279     a, b = 0, 1  
280     for i in range(n-2):  
281         a, b = b, a + b  
282     return b  
283  
284 def fib37(n):  
285     if n < 2:  
286         return n  
287     a, b = 0, 1  
288     for i in range(n-2):  
289         a, b = b, a + b  
290     return b  
291  
292 def fib38(n):  
293     if n < 2:  
294         return n  
295     a, b = 0, 1  
296     for i in range(n-2):  
297         a, b = b, a + b  
298     return b  
299  
300 def fib39(n):  
301     if n < 2:  
302         return n  
303     a, b = 0, 1  
304     for i in range(n-2):  
305         a, b = b, a + b  
306     return b  
307  
308 def fib40(n):  
309     if n < 2:  
310         return n  
311     a, b = 0, 1  
312     for i in range(n-2):  
313         a, b = b, a + b  
314     return b  
315  
316 def fib41(n):  
317     if n < 2:  
318         return n  
319     a, b = 0, 1  
320     for i in range(n-2):  
321         a, b = b, a + b  
322     return b  
323  
324 def fib42(n):  
325     if n < 2:  
326         return n  
327     a, b = 0, 1  
328     for i in range(n-2):  
329         a, b = b, a + b  
330     return b  
331  
332 def fib43(n):  
333     if n < 2:  
334         return n  
335     a, b = 0, 1  
336     for i in range(n-2):  
337         a, b = b, a + b  
338     return b  
339  
340 def fib44(n):  
341     if n < 2:  
342         return n  
343     a, b = 0, 1  
344     for i in range(n-2):  
345         a, b = b, a + b  
346     return b  
347  
348 def fib45(n):  
349     if n < 2:  
350         return n  
351     a, b = 0, 1  
352     for i in range(n-2):  
353         a, b = b, a + b  
354     return b  
355  
356 def fib46(n):  
357     if n < 2:  
358         return n  
359     a, b = 0, 1  
360     for i in range(n-2):  
361         a, b = b, a + b  
362     return b  
363  
364 def fib47(n):  
365     if n < 2:  
366         return n  
367     a, b = 0, 1  
368     for i in range(n-2):  
369         a, b = b, a + b  
370     return b  
371  
372 def fib48(n):  
373     if n < 2:  
374         return n  
375     a, b = 0, 1  
376     for i in range(n-2):  
377         a, b = b, a + b  
378     return b  
379  
380 def fib49(n):  
381     if n < 2:  
382         return n  
383     a, b = 0, 1  
384     for i in range(n-2):  
385         a, b = b, a + b  
386     return b  
387  
388 def fib50(n):  
389     if n < 2:  
390         return n  
391     a, b = 0, 1  
392     for i in range(n-2):  
393         a, b = b, a + b  
394     return b  
395  
396 def fib51(n):  
397     if n < 2:  
398         return n  
399     a, b = 0, 1  
400     for i in range(n-2):  
401         a, b = b, a + b  
402     return b  
403  
404 def fib52(n):  
405     if n < 2:  
406         return n  
407     a, b = 0, 1  
408     for i in range(n-2):  
409         a, b = b, a + b  
410     return b  
411  
412 def fib53(n):  
413     if n < 2:  
414         return n  
415     a, b = 0, 1  
416     for i in range(n-2):  
417         a, b = b, a + b  
418     return b  
419  
420 def fib54(n):  
421     if n < 2:  
422         return n  
423     a, b = 0, 1  
424     for i in range(n-2):  
425         a, b = b, a + b  
426     return b  
427  
428 def fib55(n):  
429     if n < 2:  
430         return n  
431     a, b = 0, 1  
432     for i in range(n-2):  
433         a, b = b, a + b  
434     return b  
435  
436 def fib56(n):  
437     if n < 2:  
438         return n  
439     a, b = 0, 1  
440     for i in range(n-2):  
441         a, b = b, a + b  
442     return b  
443  
444 def fib57(n):  
445     if n < 2:  
446         return n  
447     a, b = 0, 1  
448     for i in range(n-2):  
449         a, b = b, a + b  
450     return b  
451  
452 def fib58(n):  
453     if n < 2:  
454         return n  
455     a, b = 0, 1  
456     for i in range(n-2):  
457         a, b = b, a + b  
458     return b  
459  
460 def fib59(n):  
461     if n < 2:  
462         return n  
463     a, b = 0, 1  
464     for i in range(n-2):  
465         a, b = b, a + b  
466     return b  
467  
468 def fib60(n):  
469     if n < 2:  
470         return n  
471     a, b = 0, 1  
472     for i in range(n-2):  
473         a, b = b, a + b  
474     return b  
475  
476 def fib61(n):  
477     if n < 2:  
478         return n  
479     a, b = 0, 1  
480     for i in range(n-2):  
481         a, b = b, a + b  
482     return b  
483  
484 def fib62(n):  
485     if n < 2:  
486         return n  
487     a, b = 0, 1  
488     for i in range(n-2):  
489         a, b = b, a + b  
490     return b  
491  
492 def fib63(n):  
493     if n < 2:  
494         return n  
495     a, b = 0, 1  
496     for i in range(n-2):  
497         a, b = b, a + b  
498     return b  
499  
500 def fib64(n):  
501     if n < 2:  
502         return n  
503     a, b = 0, 1  
504     for i in range(n-2):  
505         a, b = b, a + b  
506     return b  
507  
508 def fib65(n):  
509     if n < 2:  
510         return n  
511     a, b = 0, 1  
512     for i in range(n-2):  
513         a, b = b, a + b  
514     return b  
515  
516 def fib66(n):  
517     if n < 2:  
518         return n  
519     a, b = 0, 1  
520     for i in range(n-2):  
521         a, b = b, a + b  
522     return b  
523  
524 def fib67(n):  
525     if n < 2:  
526         return n  
527     a, b = 0, 1  
528     for i in range(n-2):  
529         a, b = b, a + b  
530     return b  
531  
532 def fib68(n):  
533     if n < 2:  
534         return n  
535     a, b = 0, 1  
536     for i in range(n-2):  
537         a, b = b, a + b  
538     return b  
539  
540 def fib69(n):  
541     if n < 2:  
542         return n  
543     a, b = 0, 1  
544     for i in range(n-2):  
545         a, b = b, a + b  
546     return b  
547  
548 def fib70(n):  
549     if n < 2:  
550         return n  
551     a, b = 0, 1  
552     for i in range(n-2):  
553         a, b = b, a + b  
554     return b  
555  
556 def fib71(n):  
557     if n < 2:  
558         return n  
559     a, b = 0, 1  
560     for i in range(n-2):  
561         a, b = b, a + b  
562     return b  
563  
564 def fib72(n):  
565     if n < 2:  
566         return n  
567     a, b = 0, 1  
568     for i in range(n-2):  
569         a, b = b, a + b  
570     return b  
571  
572 def fib73(n):  
573     if n < 2:  
574         return n  
575     a, b = 0, 1  
576     for i in range(n-2):  
577         a, b = b, a + b  
578     return b  
579  
580 def fib74(n):  
581     if n < 2:  
582         return n  
583     a, b = 0, 1  
584     for i in range(n-2):  
585         a, b = b, a + b  
586     return b  
587  
588 def fib75(n):  
589     if n < 2:  
590         return n  
591     a, b = 0, 1  
592     for i in range(n-2):  
593         a, b = b, a + b  
594     return b  
595  
596 def fib76(n):  
597     if n < 2:  
598         return n  
599     a, b = 0, 1  
600     for i in range(n-2):  
601         a, b = b, a + b  
602     return b  
603  
604 def fib77(n):  
605     if n < 2:  
606         return n  
607     a, b = 0, 1  
608     for i in range(n-2):  
609         a, b = b, a + b  
610     return b  
611  
612 def fib78(n):  
613     if n < 2:  
614         return n  
615     a, b = 0, 1  
616     for i in range(n-2):  
617         a, b = b, a + b  
618     return b  
619  
620 def fib79(n):  
621     if n < 2:  
622         return n  
623     a, b = 0, 1  
624     for i in range(n-2):  
625         a, b = b, a + b  
626     return b  
627  
628 def fib80(n):  
629     if n < 2:  
630         return n  
631     a, b = 0, 1  
632     for i in range(n-2):  
633         a, b = b, a + b  
634     return b  
635  
636 def fib81(n):  
637     if n < 2:  
638         return n  
639     a, b = 0, 1  
640     for i in range(n-2):  
641         a, b = b, a + b  
642     return b  
643  
644 def fib82(n):  
645     if n < 2:  
646         return n  
647     a, b = 0, 1  
648     for i in range(n-2):  
649         a, b = b, a + b  
650     return b  
651  
652 def fib83(n):  
653     if n < 2:  
654         return n  
655     a, b = 0, 1  
656     for i in range(n-2):  
657         a, b = b, a + b  
658     return b  
659  
660 def fib84(n):  
661     if n < 2:  
662         return n  
663     a, b = 0, 1  
664     for i in range(n-2):  
665         a, b = b, a + b  
666     return b  
667  
668 def fib85(n):  
669     if n < 2:  
670         return n  
671     a, b = 0, 1  
672     for i in range(n-2):  
673         a, b = b, a + b  
674     return b  
675  
676 def fib86(n):  
677     if n < 2:  
678         return n  
679     a, b = 0, 1  
680     for i in range(n-2):  
681         a, b = b, a + b  
682     return b  
683  
684 def fib87(n):  
685     if n < 2:  
686         return n  
687     a, b = 0, 1  
688     for i in range(n-2):  
689         a, b = b, a + b  
690     return b  
691  
692 def fib88(n):  
693     if n < 2:  
694         return n  
695     a, b = 0, 1  
696     for i in range(n-2):  
697         a, b = b, a + b  
698     return b  
699  
700 def fib89(n):  
701     if n < 2:  
702         return n  
703     a, b = 0, 1  
704     for i in range(n-2):  
705         a, b = b, a + b  
706     return b  
707  
708 def fib90(n):  
709     if n < 2:  
710         return n  
711     a, b = 0, 1  
712     for i in range(n-2):  
713         a, b = b, a + b  
714     return b  
715  
716 def fib91(n):  
717     if n < 2:  
718         return n  
719     a, b = 0, 1  
720     for i in range(n-2):  
721         a, b = b, a + b  
722     return b  
723  
724 def fib92(n):  
725     if n < 2:  
726         return n  
727     a, b = 0, 1  
728     for i in range(n-2):  
729         a, b = b, a + b  
730     return b  
731  
732 def fib93(n):  
733     if n < 2:  
734         return n  
735     a, b = 0, 1  
736     for i in range(n-2):  
737         a, b = b, a + b  
738     return b  
739  
740 def fib94(n):  
741     if n < 2:  
742         return n  
743     a, b = 0, 1  
744     for i in range(n-2):  
745         a, b = b, a + b  
746     return b  
747  
748 def fib95(n):  
749     if n < 2:  
750         return n  
751     a, b = 0, 1  
752     for i in range(n-2):  
753         a, b = b, a + b  
754     return b  
755  
756 def fib96(n):  
757     if n < 2:  
758         return n  
759     a, b = 0, 1  
760     for i in range(n-2):  
761         a, b = b, a + b  
762     return b  
763  
764 def fib97(n):  
765     if n < 2:  
766         return n  
767     a, b = 0, 1  
768     for i in range(n-2):  
769         a, b = b, a + b  
770     return b  
771  
772 def fib98(n):  
773     if n < 2:  
774         return n  
775     a, b = 0, 1  
776     for i in range(n-2):  
777         a, b = b, a + b  
778     return b  
779  
780 def fib99(n):  
781     if n < 2:  
782         return n  
783     a, b = 0, 1  
784     for i in range(n-2):  
785         a, b = b, a + b  
786     return b  
787  
788 def fib100(n):  
789     if n < 2:  
790         return n  
791     a, b = 0, 1  
792     for i in range(n-2):  
793         a, b = b, a + b  
794     return b  
795  
796 def fib101(n):  
797     if n < 2:  
798         return n  
799     a, b = 0, 1  
800     for i in range(n-2):  
801         a, b = b, a + b  
802     return b  
803  
804 def fib102(n):  
805     if n < 2:  
806         return n  
807     a, b = 0, 1  
808     for i in range(n-2):  
809         a, b = b, a + b  
810     return b  
811  
812 def fib103(n):  
813     if n < 2:  
814         return n  
815     a, b = 0, 1  
816     for i in range(n-2):  
817         a, b = b, a + b  
818     return b  
819  
820 def fib104(n):  
821     if n < 2:  
822         return n  
823     a, b = 0, 1  
824     for i in range(n-2):  
825         a, b = b, a + b  
826     return b  
827  
828 def fib105(n):  
829     if n < 2:  
830         return n  
831     a, b = 0, 1  
832     for i in range(n-2):  
833         a, b = b, a + b  
834     return b  
835  
836 def fib106(n):  
837     if n < 2:  
838         return n  
839     a, b = 0, 1  
840     for i in range(n-2):  
841         a, b = b, a + b  
842     return b  
843  
844 def fib107(n):  
845     if n < 2:  
846         return n  
847     a, b = 0, 1  
848     for i in range(n-2):  
849         a, b = b, a + b  
850     return b  
851  
852 def fib108(n):  
853     if n < 2:  
854         return n  
855     a, b = 0, 1  
856     for i in range(n-2):  
857         a, b = b, a + b  
858     return b  
859  
860 def fib109(n):  
861     if n < 2:  
862         return n  
863     a, b = 0, 1  
864     for i in range(n-2):  
865         a, b = b, a + b  
866     return b  
867  
868 def fib110(n):  
869     if n < 2:  
870         return n  
871     a, b = 0, 1  
872     for i in range(n-2):  
873         a, b = b, a + b  
874     return b  
875  
876 def fib111(n):  
877     if n < 2:  
878         return n  
879     a, b = 0, 1  
880     for i in range(n-2):  
881         a, b = b, a + b  
882     return b  
883  
884 def fib112(n):  
885     if n < 2:  
886         return n  
887     a, b = 0, 1  
888     for i in range(n-2):  
889         a, b = b, a + b  
890     return b  
891  
892 def fib113(n):  
893     if n < 2:  
894         return n  
895     a, b = 0, 1  
896     for i in range(n-2):  
897         a, b = b, a + b  
898     return b  
899  
900 def fib114(n):  
901     if n < 2:  
902         return n  
903     a, b = 0, 1  
904     for i in range(n-2):  
905         a, b = b, a + b  
906     return b  
907  
908 def fib115(n):  
909     if n < 2:  
910         return n  
911     a, b = 0, 1  
912     for i in range(n-2):  
913         a, b = b, a + b  
914     return b  
915  
916 def fib116(n):  
917     if n < 2:  
918         return n  
919     a, b = 0, 1  
920     for i in range(n-2):  
921         a, b = b, a + b  
922     return b  
923  
924 def fib117(n):  
925     if n < 2:  
926         return n  
927     a, b = 0, 1  
928     for i in range(n-2):  
929         a, b = b, a + b  
930     return b  
931  
932 def fib118(n):  
933     if n < 2:  
934         return n  
935     a, b = 0, 1  
936     for i in range(n-2):  
937         a, b = b, a + b  
938     return b  
939  
940 def fib119(n):  
941     if n < 2:  
942         return n  
943     a, b = 0, 1  
944     for i in range(n-2):  
945         a, b = b, a + b  
946     return b  
947  
948 def fib120(n):  
949     if n < 2:  
950         return n  
951     a, b = 0, 1  
952     for i in range(n-2):  
953         a, b = b, a + b  
954     return b  
955  
956 def fib121(n):  
957     if n < 2:  
958         return n  
959     a, b = 0, 1  
960     for i in range(n-2):  
961         a, b = b, a + b  
962     return b  
963  
964 def fib122(n):  
965     if n < 2:  
966         return n  
967     a, b = 0, 1  
968     for i in range(n-2):  
969         a, b = b, a + b  
970     return b  
971  
972 def fib123(n):  
973     if n < 2:  
974         return n  
975     a, b = 0, 1  
976     for i in range(n-2):  
977         a, b = b, a + b  
978     return b  
979  
980 def fib124(n):  
981     if n < 2:  
982         return n  
983     a, b = 0, 1  
984     for i in range(n-2):  
985         a, b = b, a + b  
986     return b  
987  
988 def fib125(n):  
989     if n < 2:  
990         return n  
991     a, b = 0, 1  
992     for i in range(n-2):  
993         a, b = b, a + b  
994     return b  
995  
996 def fib126(n):  
997     if n < 2:  
998         return n  
999     a, b = 0, 1  
1000    for i in range(n-2):  
1001        a, b = b, a + b  
1002    return b  
1003  
1004 def fib127(n):  
1005     if n < 2:  
1006         return n  
1007     a, b = 0, 1  
1008     for i in range(n-2):  
1009         a, b = b, a + b  
1010     return b  
1011  
1012 def fib128(n):  
1013     if n < 2:  
1014         return n  
1015     a, b = 0, 1  
1016     for i in range(n-2):  
1017         a, b = b, a + b  
1018     return b  
1019  
1020 def fib129(n):  
1021     if n < 2:  
1022         return n  
1023     a, b = 0, 1  
1024     for i in range(n-2):  
1025         a, b = b, a + b  
1026     return b  
1027  
1028 def fib130(n):  
1029     if n < 2:  
1030         return n  
1031     a, b = 0, 1  
1032     for i in range(n-2):  
1033         a, b = b, a + b  
1034     return b  
1035  
1036 def fib131(n):  
1037     if n < 2:  
1038         return n  
1039     a, b = 0, 1  
1040     for i in range(n-2):  
1041         a, b = b, a + b  
1042     return b  
1043  
1044 def fib132(n):  
1045     if n < 2:  
1046         return n  
1047     a, b = 0, 1  
1048     for i in range(n-2):  
1049         a, b = b, a + b  
1050     return b  
1051  
1052 def fib133(n):  
1053     if n < 2:  
1054         return n  
1055     a, b = 0, 1  
1056     for i in range(n-2):  
1057         a, b = b, a + b  
1058     return b  
1059  
1060 def fib134(n):  
1061     if n < 2:  
1062         return n  
1063     a, b = 0, 1  
1064     for i in range(n-2):  
1065         a, b = b, a + b  
1066     return b  
1067  
1068 def fib135(n):  
1069     if n < 2:  
1070         return n  
1071     a, b = 0,
```

A Few Simple Optimisations Applied

- Dependency Cleanup
 - DB Drivers, Spring Boot Actuator, Jackson, Tomcat Websocket, ...
- Bellsoft Builder (musl) / Base Builder Tiny
- JLink
- JVM Parameters (java-memory-calculator)
- Application Class Data Sharing (AppCDS)
- Spring AOT
- Lazy Spring Beans
- Fixing Spring Boot Config Location
- Virtual Threads

Bellsoft Builder (musl)

```
sdk use java 21.0.10-librca

mvn spring-boot:build-image

docker run -p 8080:8080 \
  -e spring.threads.virtual.enabled=true \
  -e spring.main.lazy-initialization=true \
  -e spring.data.jpa.repositories.bootstrap-mode=lazy \
  -e spring.config.location=classpath:application.properties \
  -t spring-petclinic-optimized-builder-bellsoft-musl:4.0.0-SNAPSHOT
```

	VT	Build	Image	Startup	Initial RAM	R/s	RAM	50%	75%	90%	99%	99.9%
 Java 17		76s	277MB	4.503s	263MB	338/s	374MB	116ms	197ms	251ms	386ms	489ms
Java 17	-	53s	145MB	3.353s	224MB	249/s	363MB	188ms	294ms	409ms	734ms	1011ms
Java 21		50s	211MB	2.97s	234MB	332/s	369MB	109ms	206ms	310ms	589ms	797ms
Java 25		57s	318MB	2.933s	231MB	338/s	676MB	113ms	203ms	298ms	492ms	646ms

Base Builder Tiny

```
sdk use java 21.0.10-librca

mvn spring-boot:build-image

docker run -p 8080:8080 \
  -e spring.threads.virtual.enabled=true \
  -e spring.main.lazy-initialization=true \
  -e spring.data.redis.repositories.bootstrapMode=lazy \
```

	VT	AC	COH	Build	Image	Startup	Initial RAM	R/s	RAM	50%	75%	90%	99%	99.9%
 Java 17				76s	277MB	4.503s	263MB	338/s	374MB	116ms	197ms	251ms	386ms	489ms
Java 17				59s	311MB	3.266s	216MB	289/s	346MB	118ms	257ms	389ms	659ms	903ms
Java 21				58s	318MB	3.113s	225MB	308/s	362MB	112ms	215ms	377ms	610ms	889ms
Java 25				61s	344MB	3.238s	221MB	333/s	643MB	113ms	205ms	300ms	497ms	679ms
Java 25				61s	344MB	3.307s	225MB	332/s	646MB	113ms	204ms	300ms	498ms	681ms
Java 25				-	-	3.145s	221MB	331/s	643MB	113ms	206ms	302ms	505ms	699ms
Java 25				59s	547MB	4.034s	240MB	324/s	533MB	117ms	208ms	304ms	501ms	687ms
Java 25				-	-	3.591s	230MB	1936/s	343MB	23ms	26ms	28ms	61ms	856ms

Did I miss something? 🧐

Let me/us know! 🙋

... or get in touch later! 🙈



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